

University of Prishtina “Hasan PRISHTINA”



**Faculty of Electrical and Computer Engineering
Master in Computer and Software Engineering, 2025-2026
Course: Data Preparation and Visualization**

Week IX

Importance of data volume, similarities and non-similarities - II

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Cosine Similarity

➤ If d_1 and d_2 are two document vectors, then $\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \cdot \|d_2\|}$

where $\cdot$ indicates vector dot product and $\|d\|$ is the length of vector d .

➤ Example:

$$d_1 = 3 \ 2 \ 0 \ 5 \ 0 \ 0 \ 0 \ 2 \ 0 \ 0$$

$$d_2 = 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 2$$

$$d_1 \cdot d_2 = ?$$

$$\|d_1\| = ?$$

$$\|d_2\| = ?$$

$$\cos(d_1, d_2) = ?$$

Cosine Similarity

- If d_1 and d_2 are two document vectors, then:
- $$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{||d_1|| \cdot ||d_2||}$$
- where $\cdot$ indicates vector dot product and $||d||$ is the length of vector d .

- Example:

$$d_1 = 3 \ 2 \ 0 \ 5 \ 0 \ 0 \ 0 \ 2 \ 0 \ 0$$

$$d_2 = 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 2$$

$$d_1 \cdot d_2 = 3*1 + 2*0 + 0*0 + 5*0 + 0*0 + 0*0 + 0*0 + 2*1 + 0*0 + 0*2 = \mathbf{5}$$

$$||d_1|| = (3*3 + 2*2 + 0*0 + 5*5 + 0*0 + 0*0 + 0*0 + 2*2 + 0*0 + 0*0) ^{0.5} = (42) ^{0.5} = \mathbf{6.481}$$

$$||d_2|| = (1*1 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 0*0 + 1*1 + 0*0 + 2*2) ^{0.5} = (6) ^{0.5} = \mathbf{2.245}$$

$$\cos(d_1, d_2) = \mathbf{.3150}$$

Cosine Similarity – Class Exercise

- If d_1 and d_2 are two document vectors, then

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \cdot \|d_2\|}$$

where $\cdot$ indicates vector dot product and $\|d\|$ is the length of vector d .

- Example:

$d_1 = 4 \ 1 \ 2 \ 5 \ 2 \ 1 \ 4 \ 1 \ 3 \ 2$

$d_2 = 2 \ 0 \ 2 \ 0 \ 2 \ 1 \ 4 \ 2 \ 3 \ 3$

$d_1 \cdot d_2 = ?$

$\|d_1\| = ?$

$\|d_2\| = ?$

$\cos(d_1, d_2) = ?$

Cosine Similarity – Class Exercise

- If d_1 and d_2 are two document vectors, then

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{\|d_1\| \cdot \|d_2\|}$$

where $\cdot$ indicates vector dot product and $\|d\|$ is the length of vector d .

- Example:

$d_1 =$	6	2	4	7	1	5	8	3	9	3	5	9	8	17	6	12
$d_2 =$	13	4	8	18	2	1	4	2	10	3	17	12	11	7	15	4

$d_1 \cdot d_2 = ?$

$\|d_1\| = ?$

$\|d_2\| = ?$

$\cos(d_1, d_2) = ?$

Summary Statistics

- Summary statistics are numbers that summarize properties of the data:
 - Summarized properties include frequency, location and spread:
 - Examples: location – mean:
 - spread - standard deviation.
 - Most summary statistics can be calculated in a single pass through the data.

Frequency and Mode

- The frequency of an attribute value is the percentage of time the value occurs in the data set:
 - For example, given the attribute 'gender' and a representative population of people, the gender 'female' occurs about 50% of the time.
- The mode of an attribute is the most frequent attribute value;
- The notions of frequency and mode are typically used with categorical data.

Percentiles

- For continuous data, the notion of a percentile is more useful;

*Given an ordinal or continuous attribute x and a number p between **0** and **100**, the p^{th} percentile is a value of x such that $p\%$ of the observed values of x are less than;*

- For instance, the 51 percentile is the value such that 49% of all values of x are less than.

Measures of Location: Mean and Median

- The mean is the most common measure of the location of a set of points;
- However, the mean is very sensitive to outliers;
- Thus, the median or a trimmed mean is also commonly used.

$$\text{mean}(x) = \bar{x} = \frac{1}{m} \sum_{i=1}^m x_i$$

$$\text{median}(x) = \begin{cases} x_{(r+1)} & \text{if } m \text{ is odd, i.e., } m = 2r + 1 \\ \frac{1}{2}(x_{(r)} + x_{(r+1)}) & \text{if } m \text{ is even, i.e., } m = 2r \end{cases}$$

Measures of Spread: Range and Variance

- Range is the difference between the max and min;
- The variance or standard deviation is the most common measure of the spread of a set of points:

$$\text{variance}(x) = s_x^2 = \frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2$$

- However, this is also sensitive to outliers, so that other measures are often used:
 - AAD (Average Absolute Deviation) and MAD (Mean Absolute Deviation).

$$\text{AAD}(x) = \frac{1}{m} \sum_{i=1}^m |x_i - \bar{x}|$$

$$\text{MAD}(x) = \text{median}\left(\{|x_1 - \bar{x}|, \dots, |x_m - \bar{x}|\}\right)$$

$$\text{interquartile range}(x) = x_{75\%} - x_{25\%}$$

Measures of Spread: Range and Variance

➤ Average Absolute Deviation:

The average absolute deviation of a set $\{x_1, x_2, \dots, x_n\}$ is

$$\frac{1}{n} \sum_{i=1}^n |x_i - m(X)|.$$

➤ For example, for the dataset $\{2, 2, 3, 4, 14\}$:

Measure of central tendency $m(X)$	Average absolute deviation
Mean = 5	$\frac{ 2 - 5 + 2 - 5 + 3 - 5 + 4 - 5 + 14 - 5 }{5} = 3.6$
Median = 3	$\frac{ 2 - 3 + 2 - 3 + 3 - 3 + 4 - 3 + 14 - 3 }{5} = 2.8$
Mode = 2	$\frac{ 2 - 2 + 2 - 2 + 3 - 2 + 4 - 2 + 14 - 2 }{5} = 3.0$

Questions?

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